Lecture 14 // Oct. 8 // Lifting to $\mathbb{Z}$-grading and the Filtered Chain Complex

Last time: we defined the Irred Inst. Chain Cpx $C(4)$, which is a $\mathbb{Z}/8$ graded complex based at the reducible Θ. The chain cpx acts strangely at Θ: the grading drops by 5 for flows passing through it, so there's an issue with broken flowlines and ∂^2 isn't 0.

We can get a chain cpx. with the reducible sol'n by encoding the behavior at Θ into two maps.

$D_1 : C_1 \to \mathbb{F}$, which encodes flows into Θ

$D_2 \mathbb{F} \to C_{(-4)}$, which encodes flows out of Θ

From these we can define two chain cpx's for which $\partial^2 = 0$ by either forgetting flows into / out Θ

$$\overline{C} = C \oplus \mathbb{F}_0, \quad \overline{\partial} = \partial + D_1$$

$$\underline{C} = C \oplus \mathbb{F}_{(-3)} \quad \underline{\partial} = \partial + D_2$$

Q Why do we care about the reducible(s)?

A The cobordism maps usually come from the D_1, D_2 maps

$\mathbb{Z}$-Lifted version of IFH

We can lift the CS functional to be $\mathbb{R}$ valued (instead of $\mathbb{Z}$) by choosing a lift of $\mathcal{B}^*$ to the universal cover, $\tilde{\mathcal{B}}^*$

$$\pi_1(\mathcal{B}^*) \cong \mathbb{Z}, \quad \pi_1(\tilde{\mathcal{B}}^*) = 1$$

Recall that $\pi_1(\mathcal{B}^*) \cong \mathbb{Z}$ since we are quotienting the space of connections, which is contractible, by $\mathcal{G}$ which has $\mathbb{Z}$-many path components, one in each degree.

To get $\tilde{\mathcal{B}}^*$, we must instead quotient by by a subgroup of $\mathcal{G}$ which is $\mathbb{Z}$-times smaller, ie, choosing one path components as a "basepoint". The most obvious one to choose is

the degree 0 one.

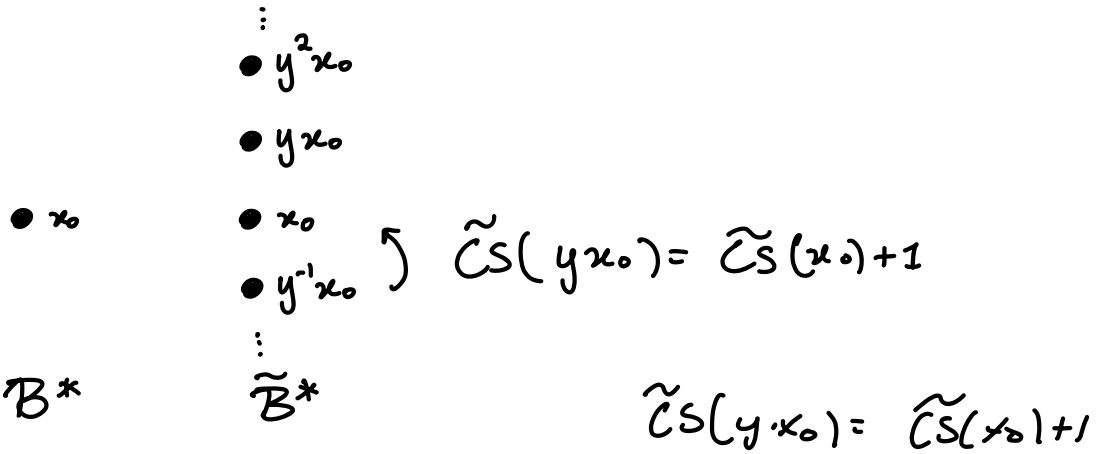
Def The based gauge group is the subgroup $\mathcal{G}_{\mathcal{J}0} \leq \mathcal{G}_{\mathcal{J}}$ of degree zero gauge transformations

$$\mathcal{G}_{\mathcal{J}0} := \{ g \in \mathcal{G}_{\mathcal{J}} \mid dg(g^{-1}) = 0 \}$$

By construction, $\pi_0(\mathcal{G}_{\mathcal{J}0}) = 1$. Like before, we define the configuration spaces $\widetilde{\mathcal{B}}$ and $\widetilde{\mathcal{B}}^*$ by

$$\widetilde{\mathcal{B}} = \mathcal{A} / \mathcal{G}_{\mathcal{J}0}$$
$$\widetilde{\mathcal{B}}^* = \mathcal{A}^{irred} / \mathcal{G}_{\mathcal{J}0}$$

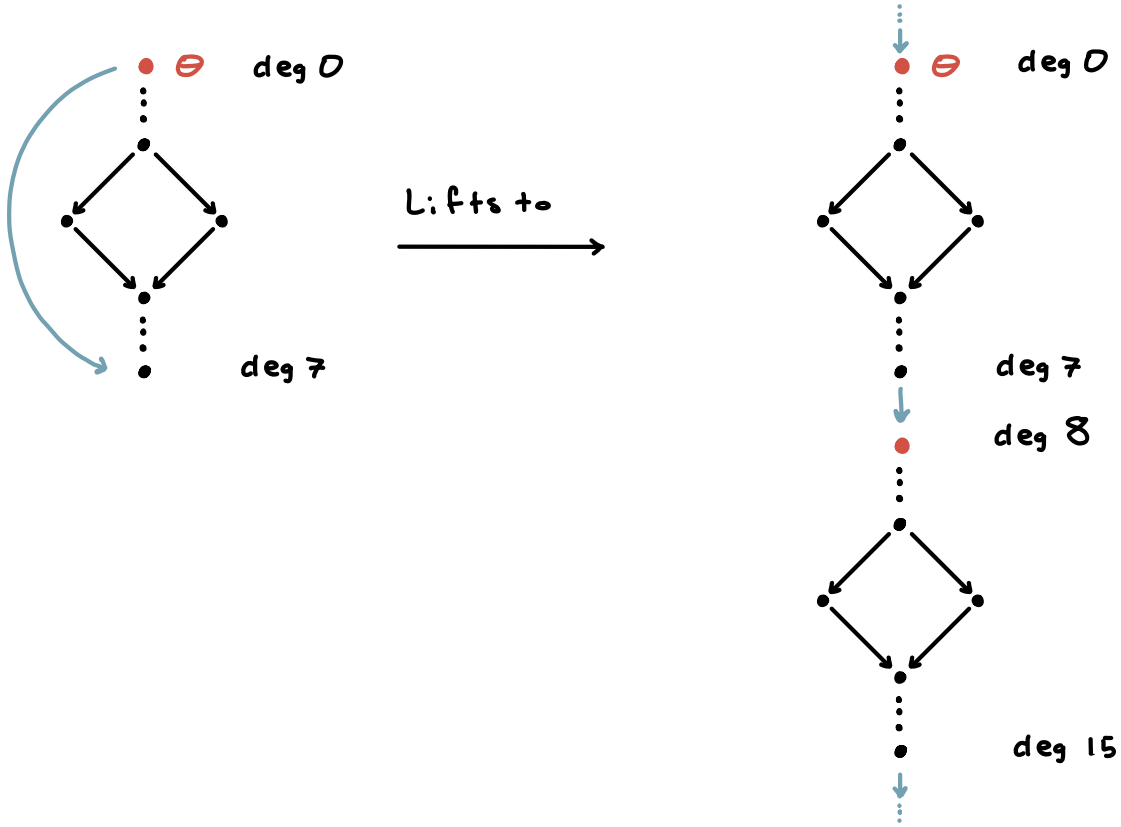
Note that $\widetilde{\mathcal{B}}^*$ is a $\mathbb{Z}:1$ cover of $\mathcal{B}^*$, so the Chern-Simons functional $CS: \mathcal{B}^* \to \mathbb{R}/\mathbb{Z}$ lifts to a $\mathbb{R}$-valued map $\widetilde{CS}: \widetilde{\mathcal{B}}^* \to \mathbb{R}$. Repeating the IFH construction, we obtain a $\mathbb{Z}$-graded complex over $\mathbb{F}[y^{\pm 1}]$, where y is in degree 8 and multiplying by y encodes switching between lifts, so increases the value of the Chern-Simons functional by one.



We will also refer to the $\mathbb{Z}$-lifted version as $C(y)$.

$$\overline{C} = C \otimes \mathbb{F}[y^{\pm 1}]_{(0)}$$

$$\underline{C} = C \otimes \mathbb{F}[y^{\pm 1}]_{(-3)}$$



The Chern-Simons Filtration

Def A $\mathbb{R}$-valued filtration on C is a collection of subcomplexes $\{C_s\}_{s\in\mathbb{R}}$ s.t. 1) $C_r \subseteq C_s$ for $r \le s$

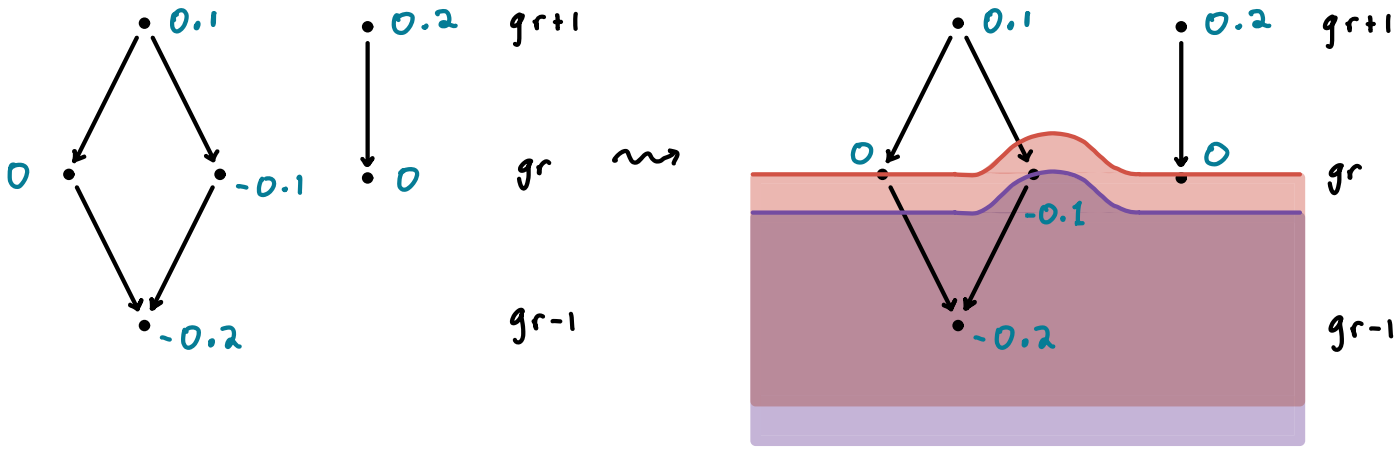
2) $\bigcup_s C_s = C$

Given a filtered cpx, we can get filtered subquotients $C^{[r,s]} = C^s/C^r$ for each pair $r<s$. Note that $[r,s]$ actually means $(r,s]$, so the generators $x \in C^{[r,s]}$ are the generators $x \in C$ with Chern-Simons value $r < \widetilde{CS}(x) \le s$.

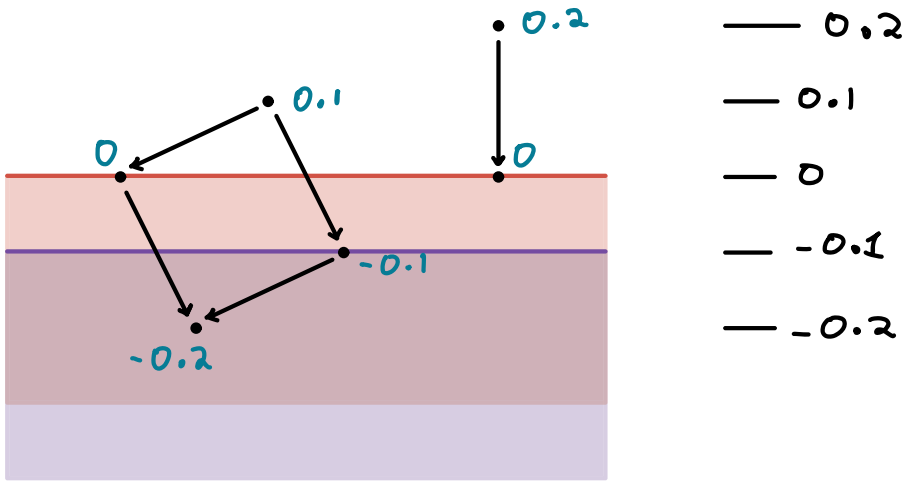
Observe that $\widetilde{CS}$ gives C the structure of a filtered complex. For each generator (critical point), we can look at the value of the (perturbed) CS-functional at each x.

Diagrammatically, the filtration looks like the following. The values of CS for each generator are written in blue, and the complex is arranged

vertically by $\mathbb{Z}$-grading.



If this complex was instead arranged vertically by Chern-Simons value, the filtration would look like horizontal lines:



Rmk The $\mathbb{Z}$-grading is indep from the cs value, but they both go down under the differential

Rmk If B^* has nontrivial π_1, then the grading shift for flows which use this π_1 is larger than one, since the expected dim for the flowline is higher. Lifting to $\widetilde{B}^*$ "unwinds" this π_1, so gradings always drop by 1.

By trivializing the bundle, we get a connection for which $\widetilde{CS}$ vanishes; we take this to be the lift of Θ.

This filtration also descends to one on $\underline{C}$ and $\overline{C}$.

Def A filtered chain map $f: C_1 \to C_2$ is a chain map which is non-increasing with respect to the filtration, ie, $f(C_1^s) \subseteq C_2^s \quad \forall s$

Def A filtered chain homotopy H between filtered chain maps f and g is a chain homotopy for which $H(C_1^s) \subset C_2^s \quad \forall s$

If H exists for f, g, f and g are filtered chain homotopic.

We claim that filtered htpy equiv class of C is an invariant of Y^3. To show this, we ask: what must we keep track of? What aux data was used in our def of $C(Y)$?

Well, the first thing we had to do for any bundle stuff was choosing a metric on Y.

The crit pts of $CS(Y)$ don't depend on g: the point is that they're inh. topological, eg, counting $SU(2)$ repn's of π_1

However, the differential does depend on g, since to even define a grad-like vf, we need a metric. Moreover, the definition uses the Hodge star! $\text{grad}_A CS = *_g F_A$

So strictly speaking, the CPX does depend on g a priori. Thus we must check that changing g doesn't change the FHEC.

There is a second thing. If CS doesn't have discrete CPs, we must perturb the CS funct to get discrete CPs.

So far, we've brushed this fact under the rug; it's actually super annoying to deal with.

A random perturbation of CS funct (or even a general Morse function) will certainly change value of $\widetilde{CS}$ at CP's. So we have no reason to think that any perturbation should play well with the CS filtration.

In reality, our defn of the IFC is incorrect. It's actually defined as a "limiting object". The idea is that we can find an arbitrarily small sequence of perturbations $\{\pi_i\}_{i=1}^{\infty}$ for which $\widetilde{CS}$ has discrete CP's.

For each perturb., we get a filtered CPX. $C^{\pi_i}(Y)$.

If we do this with enough regularity, as $\|\pi_i\| \to 0$, these complexes converge to some limiting object which we define to be $C(Y)$: $C^{\pi_i}(Y) \xrightarrow{\|\pi_i\| \to 0} C(Y)$.

However, there are many cases for which a perturb. is not needed, in which case the defn. is true as given.

For example, the Brieskorn Hz spheres $\Sigma(p,q,r)$ with three singular fibres have discrete critical points without a perturbation.

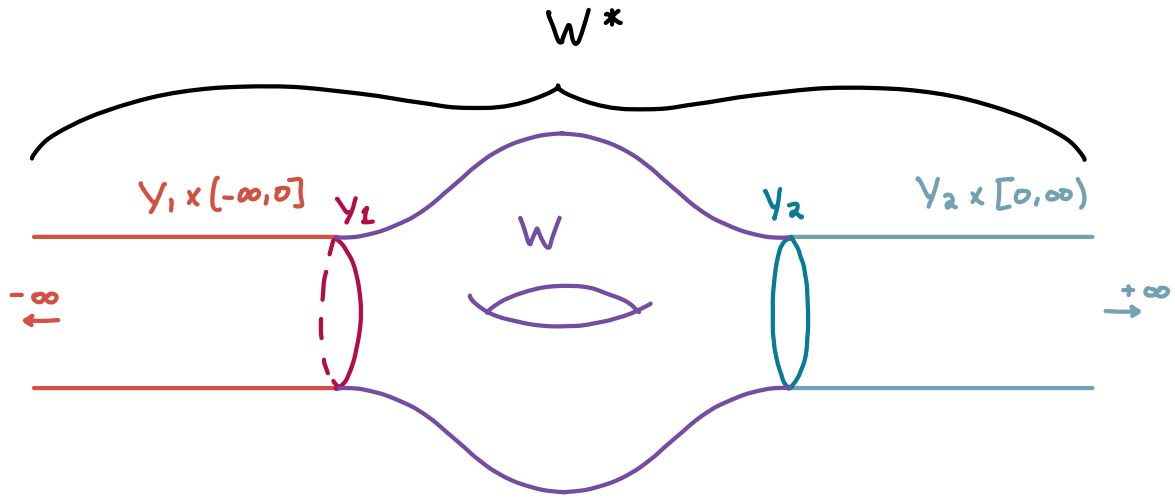
Conceptually, we'll discuss the complex as if no perturbation is needed.

Cobordism Maps

If you've seen Floer theory before, you'll have heard of the so-called TQFT structure, eg, "bordisms induce maps". This is formalized as follows

Def A Floer Homology theory F is said to have the structure of a TQFT if, for W a cob. from Y_1 to Y_2, there is a cobordism map $F_W : C(Y_1) \to C(Y_2)$

In our setting, let W be a neg-def cob. with $b_1(W) = 0$. From W, we construct the "cylindrical end mfd" W^* by attaching $Y_1 \times (-\infty, 0]$ and $-Y_2 \times [0, \infty)$ to their resp ends.



Consider the trivial bundle $W^* \times \mathbb{C}^2$.

Given $x \in C(Y_1)$ and $y \in C(Y_2)$, look at ASD connections on W^* which limit to x and y (on the left and right, respectively), modulo gauge trans on W^*.

As with ∂, space of cxns on W^* mod gauge on W^* has many diffs. top. comp's. We can think about this as follows. On Y_1 and Y_2, x and y are really just representatives of gauge equiv. classes $[x_0]$ and $[y_0]$ of cxns on Y_1 and Y_2.

If $g^* y_0 \in [y_0]$, then we also want to count connections limiting to x_0 and $g^* y_0$. As with before, we claim there is no gauge trans. on W^* sending a connection from $x_0 \to g^* y_0$ to one from $x_0 \to y_0$. In particular, if g is a gt on W^*, then $\deg(g|_{Y_1}) = \deg(g|_{Y_2})$.

Exercise Prove the claim

This shows that there are $\mathbb{Z}$-many top. classes of cxns interpolating between x_0 and y_0, determined by the difference of degree $\deg(g_1) - \deg(g_2)$ for gauge trans. g_1 on Y_1 and g_2 on Y_2.

Diagramatically, this can be seen by quotienting by a vertical $\mathbb{Z}$-shift:

